Sagnick Paul

Electrical Engineering Undergraduate - Jadavpur University

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LinkedIn — GitHub — Personal Portfolio

Profile

Electrical Engineering undergraduate focused on artificial intelligence, intelligent control systems, and robotics. Experience in embedded system design, real-time feedback control, and deep learning for classification, segmentation, and anomaly detection. Interested in applying machine learning to physical systems and autonomous platforms.

Education

Jadavpur University 2024 – 2028
B.E. in Electrical Engineering GPA: 9.24 / 10
Relevant Coursework: Control Systems, Signals & Systems, Probability & Statistics, Data Structures

Don Bosco Liluah 2010 – 2023
ISC (Class XII), Science – 92.5%

Certificates and Courses

IIT Kanpur – Professional Certificate Course in Data Science (Completed) View Certificate
IIT Delhi – Professional Certificate Program in Gen AI, ML and Intelligent Automation (Ongoing)
Simplilearn Course (Foundations: SQL) (Certified) View Certificate
Simplilearn Course (Core: Applied Data Science with Python) (Certified) View Certificate
Simplilearn Course (Core: Machine Learning) (Certified) View Certificate

Skills and Expertise

Programming — Python C Java JavaScript
Machine Learning — NumPy Pandas Scikit-Learn PyTorch TensorFlow
Embedded Systems — Arduino ESP32 L298N MPU6050 IR Sensors
Tools — Git GitHub Kaggle MATLAB Simulink

Projects

- **Brain MRI Tumor Segmentation** Github Live Demo
  CNN-based medical image segmentation with preprocessing, augmentation, Dice evaluation, and Streamlit deployment.
- **Jet Image Classification & Anomaly Detection** Github
  Built CNN classifier, applied PCA for dimensionality reduction, and developed autoencoder for anomaly detection.
- **Self-Balancing Robot**
  Designed two-wheel robot using MPU6050 and implemented PID-based real-time stabilization.
- **Employee Turnover Analytics**
  Performed EDA and developed classification models to predict employee attrition.
- **Song Cohort Analysis**
  Conducted cohort and clustering-based behavioral analysis on music consumption data.

Extra Curricular Activities

- Participated in AI/ML and robotics workshops to strengthen applied engineering skills
- Participated in a hands-on workshop on Self-Balancing Robot design involving MPU6050, Arduino, and PID-based stabilization
- Represented school in science exhibitions and technical competitions

Current Focus

- Machine Learning and Deep Learning
- Data Structures and Algorithms
- Control Systems and Optimization
- Intelligent Robotics Applications